



Project Title: Deep Learning Based Recommendation System

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Abstract

We aim to build a deep learning based recommendation system that utilizes natural language of user reviews present in the Amazon User Reviews dataset to improve the ratings predictions and to provide better recommendations.

Contents

1	Purpose of the Methodology	3
1.1	Background	3
1.2	Problem Statement	3
1.3	Objectives	4
1.3.1	Team Objectives	4
1.3.2	Individual Learning Objectives - Vivek	4
1.3.3	Individual Learning Objectives - Rutvik	4
1.3.4	Individual Learning Objectives - Sravya	5
1.4	Scope	5
2	Technical Approaches and Code UML	6
2.1	Development Environment	6
2.2	Data Collection and Preparation	6
2.3	Expected Outcome	6
2.4	Related Work	6
2.5	Implementation Details	9
3	Selection of Machine Learning Models	10
3.1	DeepCoNN	10
3.1.1	Introduction	10
3.1.2	Idea	10
3.2	HSACN	11
3.2.1	Introduction	11
3.2.2	Idea	11
3.2.3	Architecture	11
3.3	NRCMA	11
3.3.1	Introduction	11
3.3.2	Idea	12
3.3.3	Architecture	12
3.4	Final Model Selection	12
4	Model Development and Training	14
4.1	DeepCoNN Training	14
4.1.1	Hyperparameters	14

4.2	HSACN Training	15
4.2.1	Hyperparameters	15
4.3	NRCMA Training	16
4.3.1	Hyperparameters	16
5	Model Evaluation	17
5.1	Evaluation	17
6	Project Demonstration	18
6.1	Screenshots and Code Snippets	18
7	Discussion and Future Work	19
8	Conclusion	20

Chapter 1

Purpose of the Methodology

1.1 Background

Recommendation Systems are a class of machine learning algorithms that provide suggestions of items that are most relevant to users. These systems are especially useful when a user has to choose products from a large catalog. Recommendation systems are employed by online retailers like Amazon, content discovery platforms like Netflix, Prime Video, YouTube and also by other on-line platforms.

1.2 Problem Statement

Traditional methods in recommendation systems were designed to work with numerical data like ratings, however, understanding the user reviews, which are present in natural language, is important because reviews often contain valuable insights about what the user liked or disliked about the product and we can utilize this information to provide better product recommendations in the future. Deep Learning can help us process the user reviews by creating appropriate representations. Our aim is to build a deep learning based recommendation system that takes user reviews and product descriptions in account.

We choose this problem because recommendation systems are one of the most important types of machine learning solution that can help businesses directly improve sales and revenue. For instance, having a better recommendation system for an online retailer will help in providing personalized product suggestions that will eventually increase customer satisfaction and sales.

1.3 Objectives

1.3.1 Team Objectives

- Build a recommendation system that predicts user ratings for products that were not reviewed or purchased and utilize this system to present relevant products upon a search based on users past reviews and the product descriptions available for each product. This involves generating candidates using the search query and ranking them based on the ratings predicted by the deep learning model.
- Perform a literature review of the latest recommendation systems available for textual data and implement at least 4 architectures on our dataset. We would also like to compare the performance of the latest architectures, which utilize the natural language text of reviews with traditional methods like Collaborative Filtering.
- By the end of the project, we would like to understand and present the challenges in implementing real-world recommendation systems.

1.3.2 Individual Learning Objectives - Vivek

- Understand the nuances in training deep learning models in PyTorch.
- Learn about different recommendation systems algorithms and understand the use cases for each one of it.
- Implement MLOps in the project to better understand the Machine Learning workflow and the tools used in the industry.
- Gain hands-on experience deploying deep learning models to cloud services like Vertex AI.

1.3.3 Individual Learning Objectives - Rutvik

- Research about how multi-million parameter language models are hosted and trained.
- Ensure scalability is achieved through the model trained.
- Conduct an end-to-end project collaboration with peers within a dedicated time frame.
- Experiment various text embedding techniques to create optimal data for the process.

1.3.4 Individual Learning Objectives - Sravya

- Get hands-on experience with recommendation systems.
- Learn how big-data is handled and possible tools associated with it.
- Compare how different techniques work for recommendation systems (collaborative filtering vs deep CNNs)
- Implement a MLOps pipeline for project and learn how to handle the model deployment for big data.

1.4 Scope

Define the boundaries of the project, including what is and isn't covered.

Chapter 2

Technical Approaches and Code UML

2.1 Development Environment

2.2 Data Collection and Preparation

The Amazon User Reviews dataset is a large scale dataset of Amazon user reviews present in the HuggingFace Datasets library. This dataset contains rich information about user reviews and item metadata. The user reviews contain details such as rating, actual text of the review, number of helpfulness votes etc. The item metadata contains information like description, features, average ratings, title etc. There are 571M reviews posted by 54.51M users across 48.19M products and the reviews ranging from May 1996 to September 2023. The products and reviews are also grouped according to 33 different product categories such as Electronics, Health and Household, Automotive, Industrial and Scientific, etc.

This is a static dataset and does not require any access to specific APIs. This dataset was developed at the McAuley Lab at UCSD.

2.3 Expected Outcome

At the end of the project, we want to deliver a deep learning model that predicts user ratings and a web application prototype that uses the deep learning model to recommend products to the users.

2.4 Related Work

In this section, we review six recent papers introducing novel recommendation system approaches. The first two papers, HSACN and NRDPA [1], employ sim-

ilar two-tower architectures with attention mechanisms. The subsequent works explore diverse methodologies including neural content-aware collaborative filtering, aspect-based capsule networks ACAPNET [2], graph neural network applications, and BLAIR, a pretrained sentence embedding model for bridging language and items.

The HSACN (Hierarchical Self-attentive Convolution Network) paper presents a novel approach to leveraging review text for recommendation systems. The model combines convolutional neural networks (CNNs) and self-attention mechanisms in a hierarchical framework to capture both local and global text interactions. It consists of three main components: a sentence encoder, a review encoder, and a user/item encoder, which work together to learn representations at word, sentence, and review levels. The model’s key innovation is its sequence encoding module that stacks a convolution layer with a self-attention layer, allowing it to capture both local interactions (like key phrases) and global interactions. This hierarchical architecture enables flexible attention at multiple levels, focusing on important elements throughout the text. Experiments on Amazon Product Benchmark datasets demonstrate that HSACN outperforms state-of-the-art review-based recommendation models, with relative improvements ranging from 0.5 % to 2 % across different datasets. The model’s ability to effectively select informative words and sentences, considering both positive and negative sentiments in reviews, contributes to its improved performance in recommendation tasks.

In the NRDPA [1] paper, the authors introduce dynamic personalized attention mechanism at different levels i.e word and review level to model the dynamic preferences of a user for different items. For instance, a user might take into account both battery life and quality while purchasing electronics like mobile phone and a camera, however, the user may give more importance to battery life when purchasing a mobile phone and quality when purchasing a camera. So the model should take this into consideration and make required changes based on users’ preference. The authors propose personalized word level and review level attention using user’s historical rating of other items to capture user preferences and item attributes dynamically. They also introduce an interactive attention mechanism that allows passage of information between user and item encoder.

Recommender systems for e-commerce platforms are primarily based on collaborative filtering, which builds upon learning similarities between users and items from historical interaction data. These approaches inherently face the cold-start problem, as they cannot recommend novel items with no prior interaction history. Content-aware recommendation addresses this issue by incorporating textual and metadata information about items on top of collaborative filtering. However, methods in this category often rely on a shallow user/item interaction model rooted in a matrix factorization framework. In the paper, Neu-

ral content-aware collaborative filtering for cold-start music recommendation, a unified framework that alleviates these limitations and extends the recently introduced neural collaborative filtering to its content-aware counterpart is introduced. A generative model that leverages deep learning for both extracting content information from user reviews and item metadata and for modeling the interaction between user and item embeddings is proposed. The deep content feature extractor can either directly predict the item embedding or serve as a regularization prior, resulting in two variants (strict and relaxed) as shown in the paper.

Aspect-based recommendation systems focuses on identifying and utilizing specific aspects(features) of an item to generate the recommendations. By analyzing the text reviews, it gathers fine-grained sentiments attached to each of the aspects. These systems combine the importance of different aspects for individuals to make personalized recommendations based on the most relevant aspects. In the Aspect Based Capsule Network paper, the authors propose a novel architecture named ACAPNET [2] in which they utilize reverse dynamic routing process that combines users and item context features into aspect features (capsules). They use the length of the aspect feature vector as the importance of the aspect. The routing mechanism allows aspect capsules to selectively acquire information relevant to their specific aspect from the context features, avoiding homogeneity across different aspects. The authors also include mutual attention between user and item context features to model the correlations in multiple feature subspaces rather than using a single affinity matrix.

The paper, Graph Neural Networks in Recommender Systems: A Survey, discusses the impact of Graph Neural Networks for e-commerce domain using varied principles of matrix factorizations and out-of-the-box deep learning strategies. Graph Neural Networks (GNNs) are highly effective for recommender systems due to their ability to naturally model and process the graph-structured data prevalent in these systems. The paper highlights that GNNs leverage these graph structures to capture high-order connectivity patterns and aggregate both direct and indirect information about users and items through message-passing mechanisms. By iteratively updating node embeddings, GNNs enhance the expressiveness of user and item representations, leading to more accurate recommendations. It also addresses challenges like handling heterogeneous data and scalability for large-scale systems, using techniques such as edge-wise attention mechanisms and hierarchical sampling methods to manage computational costs efficiently. Moreover, the inclusion of temporal and sequential information through dynamic graph models further enriches contextual understanding, demonstrating GNNs' capability to provide a unified, flexible, and effective framework for recommendation tasks.

The BLAIR (Bridging Language and Items for Retrieval and Recommendation)

paper introduces a series of pretrained sentence embedding models designed for recommendation scenarios. Using a RoBERTa-based architecture, BLAIR encodes sentences into embeddings and is trained with a contrastive learning objective that pairs user reviews with corresponding item metadata. This approach aims to align sentence embeddings of review context and item metadata in the embedding space. The model architecture consists of a sentence encoder that learns representations from words using a word embedding layer, a sequence encoding module combining convolution and self-attention, and a sequence aggregating module using attention. BLAIR’s training objective combines a contrastive loss with an auxiliary pretraining objective (e.g., masked language modeling) to stabilize model training. The contrastive loss encourages the model to align the embeddings of related review contexts and item metadata while pushing apart unrelated pairs. By leveraging a large-scale dataset of user reviews and item metadata, BLAIR learns to bridge the gap between natural language descriptions and item representations, enabling the model to generalize across multiple domains and tasks in recommendation scenarios.

2.5 Implementation Details

Provide a step-by-step description of the coding process, algorithms implemented, and any specific methodologies followed.

- We will first start with simpler methods like Probabilistic Matrix Factorization which only takes ratings into account. The result would also act as a baseline.
- Secondly, we will train neural networks like DeepCoNN [3] which is one of the first two tower architectures that combines all reviews of a user and all the reviews of an item to learn a representation for users and items. The architecture also uses CNNs to process the text reviews. Implementing this method would give us an understanding of how to train the two tower architecture networks.
- Next, we will add attention based components such as word level and review level attention that selectively learns relevant words and reviews for a particular combination of user and item.
- In most of the two tower architectures that we reviewed the first stages process text using word embeddings & CNNs and then apply attention. We would like to test whether directly taking BERT based embeddings of the entire sentences would improve the performance as BERT based models have multiple layers of attention applied and can learn the representations of the reviews better. We would also like to train the models on different categories present in the dataset rather than just the ones mentioned in the actual papers.

Chapter 3

Selection of Machine Learning Models

This chapter introduces various methodologies implemented in creating the aforementioned Recommendation System architecture using Text Reviews. Each of the methodologies bring a unique work in creating the system.

3.1 DeepCoNN

3.1.1 Introduction

The DeepCoNN, Deep Co-Operative Neural Networks, architecture is a joint deep modelling of users and items using reviews for recommendation. This architecture specializes by introducing a two-parallel neural network architecture, also called user-tower and item-tower. These towers consider the embedded sentences for users and items.

3.1.2 Idea

The initial idea is to organize an existing dataset such that it incorporates tensors of a consistent size. The input tensor consists of user id, item id, word embedding, item embedding and rating. The architecture considers three different datasets, train, validation and test and mentions the use of batches for faster processing. A compilation of hyperparameters is mentioned in the further sections with respect to each model.

Each batch parallelly processes item reviews and user reviews in the same architecture to return a consistent size tensor which is further used to predict the rating. The parallel processing involves a look-up layer which creates the embeddings for the reviews, followed by a convolution operation, after which a max-pooling is performed and lastly there exists a fully connected layer.

3.2 HSACN

3.2.1 Introduction

HSACN aim to improve the modeling of user and item interactions by leveraging both hierarchical structures and self-attention mechanisms. This approach is particularly useful for review-based recommendation systems, where capturing the importance of different words, sentences, and reviews is crucial for accurate user and item representation.

3.2.2 Idea

The core idea behind using hierarchical self-attention networks in recommendation systems is to hierarchically model user and item representations from reviews. This involves encoding words into sentences, sentences into reviews, and reviews into user and item representations. By doing so, the network can effectively capture both local and global interactions within the text, enhancing the understanding of user preferences and item attributes.

3.2.3 Architecture

The architecture typically involves a combination of convolutional networks and self-attention. It starts with a word embedding layer, followed by a hierarchical structure of encoders:

Sentence-Level Encoder: Uses convolutional networks to extract local features from sentences. So sentence is a combo of multiple word embeddings.

Review-Level Encoder: Applies self-attention to aggregate sentence features into review representations. A review contains multiple sentence embeddings from the above layer.

User/Item-Level Encoder: Further aggregates review features using self-attention to obtain final user and item embeddings. These embeddings are then used for recommendation tasks, such as rating prediction or personalized item suggestions. This contains all the reviews the user/item gave.

This hierarchical structure allows the network to assign different weights to different parts of the reviews, highlighting the most informative aspects for recommendation purposes.

3.3 NRCMA

3.3.1 Introduction

Existing two-tower models for review-aware recommendations isolate user and item encoders, resulting in a lack of cross-modality information exchange. This

limitation hinders the ability to accurately model user-item interactions, especially in cases where reviews and ratings are inconsistent due to varying item attributes.

3.3.2 Idea

The authors propose a novel model called Neural Recommendation with Cross-Modality Mutual Attention (NRCMA). NRCMA bridges the user encoder and item encoder by employing cross-modality mutual attention mechanisms at both word and review levels. This approach enables the model to select informative words and reviews that contribute to better representation learning for users and items. The model tightly couples the two encoders, allowing them to influence each other during training in an end-to-end manner.

3.3.3 Architecture

The architecture consists of four main layers: Embedding Layer: User and item IDs are initialized as one-hot vectors and mapped into low-dimensional embeddings. Reviews are embedded using pre-trained Glove word embeddings, followed by convolutional operations to extract contextual features.

Word-Level Attention Layer: A cross-modality mutual attention mechanism is applied, where the item embedding acts as a query for the user encoder (and vice versa for the item encoder). This mechanism assigns attention weights to words in reviews, highlighting those most relevant to the target user-item interaction.

Review-Level Attention Layer: After aggregating word-level features, another cross-modality mutual attention mechanism is used to identify and emphasize the most informative reviews. This allows the model to focus on reviews that are most indicative of user preferences or item attributes.

Prediction Layer: The final user and item embeddings (obtained after attention mechanisms) are concatenated and passed through a Factorization Machine (FM) to predict ratings. The FM captures both linear and second-order interactions between features.

This design facilitates effective information exchange between reviews and ratings, improving representation learning for recommendation tasks.

3.4 Final Model Selection

The final model selection depends on the RMSE evaluation metric. Based on the batches created for each model, the model returns the average loss on the test set or the mean square error of the ratings. Thus, this will be one of the primary indicators in deciding the final model.

The next indicator that is being targeted is the training time for each of the models, as the texts are significantly north of 30GB for any average filled cate-

gory. It is crucial to identify the training time as the next best indicator.

Different models were tested because of their innovativeness to the related work. The DeepCoNN architecture is the first of its kind to introduce a parallelly running neural network which will simultaneously consider user reviews and item reviews. Previously, the deep learning architectures used to concatenate the reviews, irrespective of the user/item level in order to predict the rating. However, DeepCoNN implements a same but parallel architecture for the user reviews and item reviews which allow different review lengths to be considered for the analysis. Further, the HSACN and NRCMA are built on top of the DeepCoNN architecture which try to improve the performance of the recommendation system.

Chapter 4

Model Development and Training

4.1 DeepCoNN Training

4.1.1 Hyperparameters

The DeepCoNN architecture is trained on the following hyperparameters.

- Maximum User Review Length: 200 words
- Maximum Item Review Length: 1000 words
- Embedding Dimension: 300
- Kernel Width for Convolution Operation: [3, 5]
- Kernel Depth for Convolution Operation: 100
- Latent Factors: 50
- Factors in Factorization Machine: 10
- Batch Size: 32
- Learning Rate: 0.001

The user and item review lengths are decided by looking at the distribution of number of words per review for the user and the item. In order to make this architecture successful by not introducing any kind of bias in predictions, the architecture does not consider the current review when grouping the reviews for each user as well as item.

The dataset was already available from the source as train, test and validation splits. It was made sure that the user-item combination only occur once in a dataset.

In order to handle overfitting, a regularization lambda of 0.002 was used to prevent the model from being overfit. However, training time for a batch size of 32 and 380,000 rows which imply approximately 12,000 training batches are present in 1 epoch. Thus, one epoch takes approximately 40-50 minutes to run and the plan is to train the model for at least 10 epochs. Kaplan’s scaling law suggests that doubling the model size will improve the model performance upto a point than doubling the training data size. Due to that reason, it is ideal to keep the training size as it is and play with the hyperparameters.

The training time per epoch can be reduced by creating a custom embedding layer. According to the dataset in use, the unique number of tokens are approximately 500,000. However, the pre-trained embeddings currently in use have 3,000,000 words in the vocabulary which adds to the embedding creation time. However, introduction of a custom embedding layer in the architecture will prevent the model to consume unnecessary memory with the target individual epoch time being 10-15 minutes, which is 75% faster than the current epoch time.

4.2 HSACN Training

4.2.1 Hyperparameters

The HSACN model is trained on the following hyperparameters:

- Word embedding dimension: 300
- ID embedding dimension: 32
- Number of CNN filters: 128
- CNN window size: 3
- Review length (number of words): 12
- Sentences per review: 4
- Maximum Number of Reviews considered for each user: 3
- Maximum Number of Reviews considered for each product: 35
- Num attention heads: 4

The words per sentence, sentences per review, and reviews per user/item are based off of the analysis we conducted on the whole dataset. If any data point has less no. of aspects than these parameters, we would pad to standardize all the input representations.

The training parameters such as Learning rate, Batch size, and epochs are still to be experimented with using grid search. It's a large dataset so we're continuously optimizing our code for logging and experimentation.

4.3 NRCMA Training

4.3.1 Hyperparameters

The NRCMA model is trained on the following hyperparameters.

- Word embedding dimension: 300
- ID embedding dimension: 32
- Number of CNN filters: 80
- CNN window size: 3
- Attention vector dimension: 80
- Maximum Number of Reviews considered for each user: 3
- Maximum Number of Reviews considered for each product: 35
- Review length (number of words): 12

As all the inputs to the model have to be of same size we have to standardize the number of reviews for each user and item. After looking at the distributions of number of reviews for users and products we fixed the number of reviews for a user to be 3 and number of reviews for a product to be 35. If a user or product have less reviews, we pad the inputs with zeros. For further tuning we would employ grid searching technique.

Chapter 5

Model Evaluation

5.1 Evaluation

All the three models will be evaluated on the Root Mean Squared Error(RMSE) metric. Also, the training and inference time will be compared between all the models.

We would also compare how each model is performing across different rating values. The affect of the sparsity of the data can be understood by measuring the performance across different users or products having less than the maximum number of reviews.

Chapter 6

Project Demonstration

6.1 Screenshots and Code Snippets

Include visual evidence of the project's outputs here. Add screenshots and code snippets as needed.

Chapter 7

Discussion and Future Work

Interpret the results, discuss implications and significance, and address any limitations encountered. Suggest future work or improvements based on findings.

Chapter 8

Conclusion

Summarize the key findings, the extent to which the project objectives were achieved, and the overall significance of the project.

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